Dynamic Disequilibrium: An Agent-Based Mechanism for Bubbles without Biased Agents

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Abstract

We present the *Estimated Dynamic Equilibrium Model* (EDEM), an agent-based framework in which supply and demand are realisations of a coupled stochastic process driven by heterogeneous, error-prone agent valuations. The framework's central technical contribution is a generative mechanism for persistent disequilibrium: when market-clearing prices are repeatedly selected from the upper tail of noisy bid distributions and fed back into future valuations, expected prices drift upward even with strictly zero-mean estimation errors. We derive the bias in closed form for a clean special case ($\sigma(n-1)/(n+1)$ for n iid uniform bids, which is the classical winner's curse) and show via simulation that compounding the bias across epochs produces exponential price growth without any behavioural assumption about investor optimism. EDEM extends Miller [1977]'s divergence-of-opinion theory to a fully dynamic setting and recovers Walrasian equilibrium and Miller's static premium as limiting cases. Eight controlled experiments in the Python ABM framework Mesa, on a 32×32 real-estate neighbourhood, plus a 30-cell sensitivity grid, demonstrate six qualitatively distinct regimes—all reachable from the same agent ruleset. The result has direct implications for machine-learning valuation algorithms (Zestimate-style automated valuation models): trained on historical clearing prices, such systems are by construction fitting an upper-order statistic of plausible sale prices rather than fair value, and at deployment time function as positively-biased point estimators.

1 Introduction

The Efficient Market Hypothesis remains the dominant theory of price formation. It predicts when prices reflect available information but does not provide a generative microstructural account of when and why they fail to: bubbles, crashes, persistent cross-sectional return anomalies, and the stylised observation that real markets very rarely sit at the textbook supply–demand fixed point. Decades of empirical work have produced rival theories whose claims often contradict one another, because abundant and noisy financial data can be mined to support nearly any hypothesis [Moffitt, 2017].

The central technical claim of this paper is that two structural features—max-bid clearing and per-epoch feedback of clearing prices into agent valuations—generate persistent positive drift in realised market prices even when individual agents' estimation errors are exactly zero-mean. The mechanism is order-statistic bias. Each seller selects the maximum of multiple noisy bids, and the maximum of n zero-mean draws sits, in expectation, strictly above their mean; the realised maximum then feeds back into the next round's anchor value, and the bias compounds multiplicatively. No behavioural assumption about investor optimism, risk aversion, or attention is required.

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Thaler [1988] called this phenomenon the winner's curse and framed it as a behavioural failure: rational bid shading should, in principle, correct the expected payment. Our point is that the curse is an order-statistic effect before it is a behavioural one, and that iterating it—feeding each round's clearing price into the next round's anchor—turns a one-period bias into unbounded drift in market valuations.

The implication for machine-learning valuation matters for the NeurIPS audience. An ML valuation algorithm trained on historical clearing prices—automated valuation models (AVMs), iBuying pricing models, comparable-sale algorithms—is not fitting fair value. By construction it is fitting the upper-order statistic of plausible sale prices, because the clearing price is the maximum of the bids received. At deployment time it functions as a positively-biased point estimator that may itself provide the market with a coordinating signal pulling realised winning bids further upward.

Contributions.

1. An order-statistic mechanism for bubbles without biased agents (Section 4): in closed form, for a clean special case, the expected winning bid exceeds the home value by $\sigma(n-1)/(n+1)$ when n bidders draw zero-mean errors with dispersion σ. Compounded across epochs this produces exponential price drift.
2. A unified framework (Section 3) that nests Walrasian equilibrium and Miller's static premium as limiting cases, and admits a family of out-of-equilibrium regimes (business cycles, persistent overshoots and undershoots, runaway bubbles, constant transition) parameterised by estimation noise, balancer strength, and time-on-market.
3. Eight controlled experiments plus a 30-cell sensitivity grid (Section 6) demonstrating that the order-statistic drift produces price drift above textbook equilibrium in 100% of parameter combinations tested.
4. A discussion of implications for ML valuation systems (Section 7) connecting the model to selection-aware estimation procedures that AVM deployers should adopt.

2 Background and Related Work

Miller [1977] proposed an elegant static argument: of K investors with heterogeneous beliefs, the N most optimistic end up holding the N available shares, so the clearing price reflects the marginal optimist rather than the mean valuation. Greater divergence of opinion thus produces a price *premium*, and (because future returns are earned against the elevated price) lower subsequent returns. The half-century of empirical literature that grew up around this claim has not converged. One camp argues divergence of opinion proxies risk and predicts *higher* future returns [Williams, 1977, Mayshar, 1983, Varian, 1985, Merton, 1987, Epstein and Wang, 1994]; another finds the opposite [Doukas et al., 2006]; Moffitt [2017] summarises the impasse. The empirical record is messy, and we will argue this is exactly what one should expect.

The static framing is the source of the empirical confusion. Miller's model specifies a one-shot relationship between beliefs and the market-clearing price. Real markets are continuous processes in which today's price feeds into tomorrow's beliefs. We show that the same upper-tail-selection mechanism that produces Miller's premium (and, separately, the winner's curse [Thaler, 1988] in common-value auctions) generates a *dynamic* drift when iterated—and that the same underlying mechanism can produce either a premium or a discount in finite samples, which explains why empirical work has found both.

Closely related work in agent-based economics has used ABMs to study housing markets [Baptista et al., 2016, Geanakoplos et al., 2012] and behavioural finance generally [Shiller, 2003]. The contribution here is the structural identification of the selection mechanism, formalised analytically and validated empirically across a parameter sweep.

3 The EDEM Framework

A market is staged on a finite torus G of homes, with $Q_s(t)$ sellers $\mathcal{S}_t$ (one per home, immobile) and $Q_d(t)$ buyers $\mathcal{B}_t$ (mobile, walking the torus). Each home h has a fixed unobservable fair value $v^*(h)$

and a market value $v_t(h)$ that evolves. Each agent i valuing home h at tick t draws an estimation

$$E_{i,t}(h) = v_t(h)\,(1+\varepsilon_{i,t,h}), \qquad \varepsilon_{i,t,h} \overset{\text{iid}}{\sim} F_i(\sigma_i), \tag{1}$$

with $\mathbb{E}[\varepsilon] = 0$. The three-way index is essential: each *bid* is an independent draw. Throughout, σ_i is expressed as a decimal (0.05 means $\pm 5\%$ maximum error); parameter tables report percentages.

A buyer that lands on a seller's home posts a bid $\beta^{(t)}_{b,s} = E_{b,t}(h(s))$. Sellers accumulate bids; on patience timeout, the seller offers to sign with the buyer holding the best bid. The buyer commits iff

$$\beta \;\geq\; \frac{1}{|\mathcal{B}_b|} \sum_{\beta' \in \mathcal{B}_b} \beta', \tag{2}$$

where $\mathcal{B}_b$ is the buyer's outstanding bid set (including β). Eq. (2) is order-statistic selection on the buyer's side: committing only when the offered bid is at or above the buyer's running benchmark restricts purchases to the upper tail of the buyer's own bid distribution.

The realised market price is the output of a clearing functional $p_t = A(X_t; \theta)$ over the full market state X_t. We study two instantiations. In the discrete-event *DE* variant, A is the rolling average of the last W completed sales:

$$p_t = \frac{1}{m_t} \sum_{j=k(t)-m_t+1}^{k(t)} P^{\text{sale}}_j, \qquad m_t = \min(W, k(t)), \tag{3}$$

where $k(t)$ counts completed sales up to tick t. In the speculative *EDEM* variant there are no realised sales; instead each home's value updates multiplicatively at the end of every epoch of T ticks:

$$v_{t+T}(h) = v_t(h) \cdot \bar{r}_t, \quad \bar{r}_t = \frac{1}{|\mathcal{Y}_t|} \sum_{b \in \mathcal{Y}_t} \min_{s \in \mathcal{W}_b(t)} \frac{\beta^{(t)}_{b,s}}{v_t(h(s))} \tag{4}$$

when the set $\mathcal{Y}_t$ of buyers holding at least one winning bid is non-empty (else $\bar{r}_t = 1$). $\mathcal{W}_b(t)$ is the set of sellers on which buyer b holds the winning bid.

The supply and demand quantities are themselves integer-valued random walks driven by the price increment $\Delta p_{t-1} = p_{t-1} - p_{t-2}$:

$$Q_s(t) = Q_s(t-1) + Z^s_t, \quad \mathbb{E}[Z^s_t] = C_b \,\mathrm{sgn}(\Delta p_{t-1}). \tag{5}$$

With $C_b > 0$ rising prices add sellers (mean-reverting); with $C_b < 0$ rising prices remove sellers (trend-following); with $C_b = 0$ the population is frozen. A symmetric equation governs Q_d. A population floor of one agent per side truncates the random walk in extreme regimes.

Setting $\sigma_i \equiv 0$ collapses Eq. (1) to universal agreement; the order-statistic drift vanishes and EDEM approaches the Walrasian fixed-point equilibrium. Retaining $\sigma_i > 0$ but freezing the dynamics ($T \to \infty$, single period, no value update) recovers a Miller-style static premium mechanism.

4 The Order-Statistic Mechanism

A concrete number first. With just two bidders ($n = 2$) drawing zero- mean uniform errors of width $\sigma = 0.15$, the expected winning bid sits 5% above fair value—per home, per epoch. Iterate that for 150 epochs and the implied compound is enormous. The closed form is what the rest of this section derives.

Closed-form bias. Consider a seller s receiving $n \geq 2$ bids $\beta_i = v_t(h(s))(1+\varepsilon_i)$ with $\varepsilon_i \sim \mathcal{U}[-\sigma, +\sigma]$ iid. The expected winning bid is

$$\mathbb{E}\left[\frac{\max_i \beta_i}{v_t(h(s))}\right] = 1 + \mathbb{E}\Big[\max_i \varepsilon_i\Big] = 1 + \sigma\,\frac{n-1}{n+1}, \tag{6}$$

using the order-statistic identity for the maximum of n iid uniform variates. The right-hand side strictly exceeds 1 for all $n \geq 2$ and $\sigma > 0$; the gap—whether one calls it excess return or mispricing—grows in both n and σ.

Relation to the winner's curse. Eq. (6) recovers the classical winner's curse as a static special case. In a common-value auction, the winning bidder is selected precisely because her signal is the most favourable draw among the bidder pool. Thaler [1988] treats this as a behavioural failure; rational bid shading should correct the expected payment in any one round. Whether shading fully corrects, or only partially, is downstream of the structural fact: the winning signal comes from the upper tail, full stop. EDEM's contribution is to compound that fact—one round's clearing price becomes the next round's anchor—and to show that iteration alone turns a one-period bias into unbounded drift.

What Eq. (6) **does and does not prove.** Eq. (6) establishes that the seller-side maximum has expected ratio strictly above 1. The per-epoch update Eq. (4) compounds a related but distinct quantity $\bar{r}_t$ (the buyer-side mean of $\min_s$ winning ratios), for which we have no closed form. Two further care points apply when moving from the theorem to sample-path behaviour: *(i)* for an iid multiplicative process $v_T = v_0 \prod_t r_t$, the expectation $\mathbb{E}[v_T] = v_0\,(\mathbb{E}[r])^T$ grows multiplicatively whenever $\mathbb{E}[r] > 1$, but the median is governed by $\mathbb{E}[\log r]$; by Jensen's inequality $\mathbb{E}[r] > 1$ does not on its own imply $\mathbb{E}[\log r] > 0$. *(ii)* The fallback $\bar{r}_t = 1$ for empty-winner epochs puts a point mass at unity that the clean theorem does not capture.

Empirical bridge. Table 1 reports the distribution of $\bar{r}_t$ extracted directly from our EDEM-regime simulations, with $\bar{r}_t = v_{t+T}(h)/v_t(h)$ at every epoch boundary across all seeds. The growth condition $\mathbb{E}[\log \bar{r}_t] > 0$ holds in every regime, certifying sample-path exponential growth and closing the gap between Eq. (6) and the simulated buyer-min update.

Table 1: Empirical distribution of $\bar{r}_t$ across 1500 epochs per regime (10 seeds, 150 epochs/seed). The growth condition $\mathbb{E}[\log \bar{r}_t] > 0$ holds in every regime examined.

Regime	$\mathbb{E}[\bar{r}_t]$	$\Pr(\bar{r}_t > 1)$	$\mathbb{E}[\log \bar{r}_t]$	median $\bar{r}_t$
Run 6 ($C_b = 0, \bar{\sigma} = 0.15$)	1.0429	83.5%	+0.0411	1.044
Run 7 ($C_b = +1, \bar{\sigma} = 0.15$)	1.0104	19.7%	+0.0098	1.000
Run 7 ($C_b = -1, \bar{\sigma} = 0.15$)	1.0153	25.4%	+0.0144	1.000
Run 8 ($C_b = -1, \bar{\sigma}$ growing)	1.0287	24.9%	+0.0236	1.000

5 Implementation

EDEM is implemented in the Python ABM framework Mesa [Masad and Kazil, 2015] on a 32×32 toroidal `MultiGrid`, matching the wrapped patch world of the original NetLogo prototypes [Wilensky, 1999] from which this work descends. Per-cell state ($v_t(h)$, $v^*(h)$, last sale price, last sale tick) is a lightweight dataclass attached to each grid cell; buyers and sellers are `mesa.Agent` subclasses. Bids are mirrored dictionaries on each side. The rolling 25-sale market price of Eq. (3) is a `deque(maxlen = 25)`. The per-epoch update in Eq. (4) fires on a cycle-counter trick one tick before buyers reset, preserving their winning-bid flags. The balancer Eq. (5) fires $\lfloor |C_b| \rfloor$ deterministic swaps per epoch plus one Bernoulli swap with probability $|C_b| - \lfloor |C_b| \rfloor$; sign of C_b controls direction. A population floor of one agent per side prevents the balancer from killing the last agent.

The Mesa core lives in six modules under `edem/` totalling 973 lines including docstrings; the 42 `pytest` unit tests sit in five files of similar density. Each experiment script seeds the model deterministically and runs 10 seeds. Total reproduction time on a 2024-era laptop is approximately 25 minutes (CPU only). Code and data are released under MIT licence at a URL provided in the camera-ready version of this paper.

6 Experiments and Results

Eight controlled experiments grouped into two regimes (DE: Runs 1–5, discrete-event sales; speculative-market EDEM: Runs 6–8, multiplicative value updates) plus a 30-cell sensitivity grid (Run 9). Each experiment runs 10 seeds; figures plot the median across seeds with the 10th–90th percentile band. Full parameter tables are in Appendix B; per-run figures for runs not shown inline here are in Appendix C.

The six regimes. Three parameter axes ($\bar{\sigma}$, seller patience, C_b) and the same agent ruleset generate six qualitatively distinct steady states: band-stable, business-cycle, persistent overshoot, persistent undershoot, runaway bubble, and constant transition. Table 2 consolidates the end-of-run statistics; we flag the one entry below the table that drives the rest of this section.

Table 2: Eight EDEM runs: parameters and observed outcomes (median across ≥ 8 seeds). DE runs report median deviation of the rolling market price from the textbook equilibrium; EDEM runs report v_t/v^*.

Run	Manipulated parameter	Outcome metric	Regime
1	$\bar{\sigma} = 0.05$, patience ≤ 50	-2% to $+11\%$ band	band-stable
2	$\bar{\sigma} = 0.25$	-17% to $+34\%$ periodic	business-cycle
3	patience ≤ 100	$+14\%$ persistent	overshoot
4	demand intercept -50	-18% persistent	undershoot
5	shock at $t = 3000$, then re-shocks	moving-target chase	transitional
6	EDEM, $C_b = 0$, $\bar{\sigma} = 0.15$	$v_t/v^* = 421\times$	runaway bubble
7	EDEM, $C_b \in \{-1, 0, +1\}$	$7.7, 421, 4.2\times$ resp.	balancer-shaped
8	EDEM, $C_b = -1$, $\bar{\sigma}$ growing	$v_t/v^* = 30\times$	super-linear

Bubble regime under no balancer. With balancer disabled ($C_b = 0$) and $\bar{\sigma} = 0.15$, the market exhibits a clean log-linear bubble: a median multiplicative gain of $\sim 421\times$ over 3000 ticks, with a tight band (Fig. 1). The mechanism is exactly that of Section 4—each epoch's average winning ratio $\bar{r}_t$ is > 1 in expectation, and Eq. (4) compounds it multiplicatively. $C_b = 0$ admits all of the upward bid-selection bias unfiltered, which is exactly what this configuration is meant to expose.

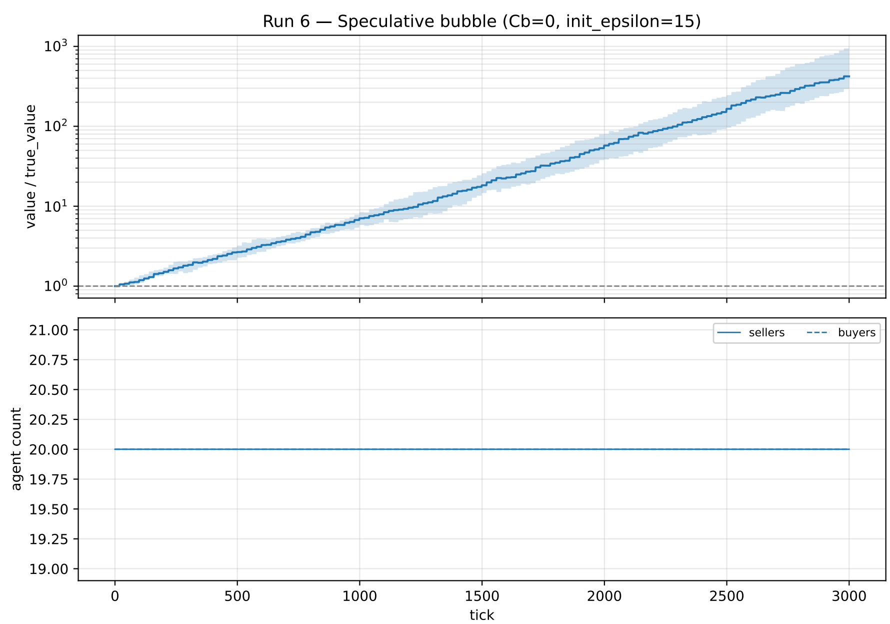

Figure 1: Run 6 (bubble, $C_b = 0$, $\bar{\sigma} = 0.15$). Top: median v_t/v^* (log scale) with 10–90 percentile band. Bottom: agent counts (frozen at 20 each because $C_b = 0$).

Balancer-coefficient sweep. Varying $C_b \in \{+1, 0, -1\}$ at fixed $\bar{\sigma} = 0.15$ yields three qualitatively distinct trajectories (Fig. 2). The mean-reverting case ($C_b = +1$) plateaus near $4\times$. The trend-following case ($C_b = -1$) reaches $\sim 7.7\times$. The no-balancer case dominates both at over $100\times$ either, ending near $421\times$. The intuitive guess, dating to earlier informal write-ups in this lineage, is that trend-following should be the most explosive regime—there is more aggregate bidding pressure when buyers swarm a shrinking pool of sellers. In the Mesa simulation the opposite holds. The trend-following balancer drains sellers to a single agent, and inflation needs sellers to inflate; the no-balancer case keeps populations balanced and admits the bid-selection bias undamped. The population panel of Fig. 2 makes the mechanism visible: the $C_b = \pm 1$ trajectories collapse into a one-vs-many state within a few hundred ticks, while $C_b = 0$ holds at the $20/20$ initialisation throughout. No setting of C_b in this grid restores $v_t/v^* = 1$.

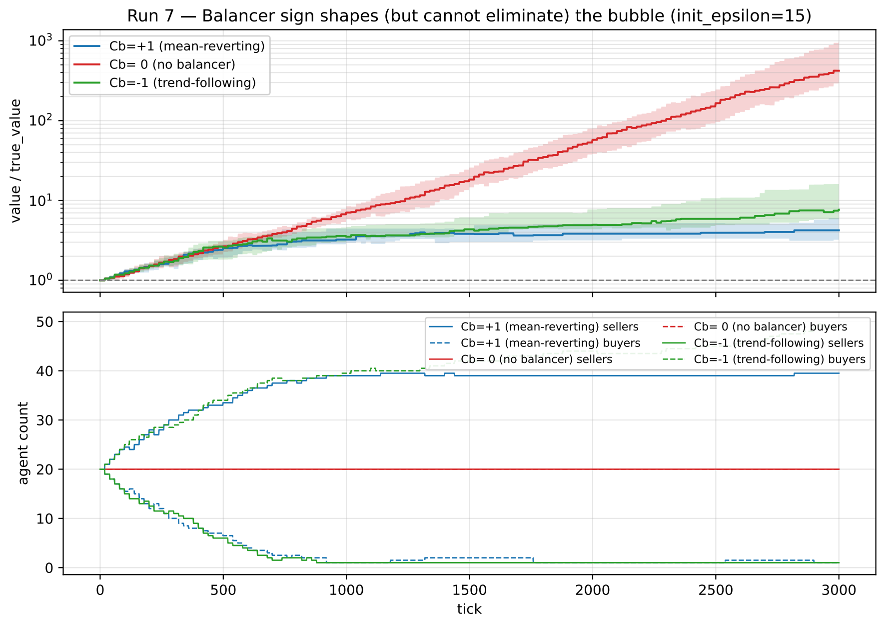

Figure 2: Run 7 (balancer sweep at $\bar{\sigma} = 0.15$). The three regimes are overlaid; $C_b = 0$ produces the strongest bubble. The population panel shows the one-vs-many state the strong balancers induce.

Sensitivity to C_b and $\bar{\sigma}$. A 5×6 grid sweeps $C_b \in \{-1, -0.5, 0, +0.5, +1\}$ and $\bar{\sigma} \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$, with 5 seeds per cell at 1500 ticks (Fig. 3). Two patterns emerge. First, the gap widens monotonically with $\bar{\sigma}$—larger divergence of opinion produces a larger multiplier, in every column. Second, every cell of the grid has $v_t/v^* > 1$, and the $C_b = 0$ column dominates each row. Of the 30 cells, 13 exceed $10\times$ within 1500 ticks. The order-statistic drift is universal across the parameter grid we explored, not a corner case.

Other regimes. DE Runs 1–5 produce the band-stable, business-cycle, overshoot, undershoot, and transitional regimes listed in Table 2. Run 4 (low agent density): mean realised price $p \approx 41$ versus textbook $p^* = 50$, an 18% shortfall, with episodes reaching $p \approx 37$ (26% at the trough). Run 5 (shocked market): the realised price never settles when demand is toggled every 2000 ticks. The figures for these runs are in Appendix C.

7 Discussion: Implications for ML Valuation

Bid-selection asymmetry, not estimation bias, drives bubbles. The estimation function Eq. (1) has zero mean error: an agent's expected valuation is exactly the home's current value. Yet the EDEM speculative regime produces unbounded upward drift even with this unbiased estimator. The mechanism is not behavioural bias but a statistical asymmetry: each seller selects the maximum bid received, and the maximum of N noisy estimates of a fair value is, in expectation, larger than the fair value itself. Eq. (4) compounds this asymmetry multiplicatively across epochs. This is the dynamic generalisation of the classical winner's curse [Thaler, 1988].

Implication for ML valuation systems. The same mechanism operates in any pricing system whose inputs are anchored to recent market-clearing prices. A machine-learning valuation algorithm trained on historical sale prices is, by construction, fitting the upper-order statistic of plausible sale prices for a similar asset—the maximum of the relevant noisy estimates—not their mean. Deployed at

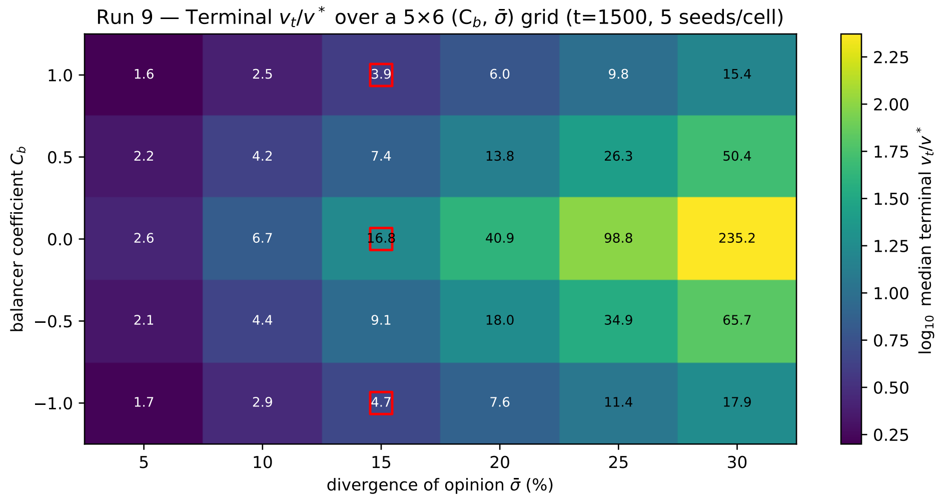

Figure 3: Run 9 (sensitivity heatmap). Median terminal v_t/v^* over a 5×6 (C_b, $\bar{\sigma}$) grid, log colour scale. Red squares mark cells corresponding to Runs 6 and 7. Every cell exceeds 1.0; 43% exceed 10.0.

scale, such an algorithm functions as a positively-biased point estimator that may provide the market with a coordinating signal pulling realised winning bids further upward, feeding back into the next training cycle.

The fix is structural, not a hyperparameter knob. A defensible deployment would predict the median rather than the mean of plausible sale prices, use a censored-data likelihood that accounts for the offer-acceptance threshold, or substitute an explicitly rank-aware estimator that targets fair value instead of winning-bid value. The 2021 wind-down of Zillow's algorithmic-iBuying programme [Zillow Group, Inc., 2021] is one episode in which a clearing-price-anchored valuation system at scale incurred substantial losses; the public record is consistent with the mechanism described here, though we do not advance EDEM as a complete account of that specific event.

Why empirical premium/discount studies disagree. Each empirical sample of the divergence-of-opinion premium is a finite-window observation along a coupled stochastic process governed by σ and C_b. Two studies sampling distinct positions on distinct trajectories may both correctly observe opposite signs. The unifying claim from EDEM is not that one camp is right and the other wrong, but that the static framing has been asking the wrong question; the relevant quantity is the trajectory of $\bar{r}_t$, not its sign at a snapshot.

8 Limitations and Conclusion

EDEM models a single asset class with no leverage, no macroeconomic environment, and non-adaptive (non-learning) agents. The 30-cell sensitivity grid covers a defensible range of $(C_b, \bar{\sigma})$ but does not explore extreme $|C_b|$ values where the population dynamics qualitatively change. Run 5's shock-recovery scenario uses a 12,000-tick budget that is too short for the market to reach its post-shock floor before the patience boost fires. Bayesian-belief and reinforcement-learning extensions of the agent rule set (Appendix D) are sketched but not implemented.

The contribution we believe travels beyond the EDEM setting is the mechanism. Bubbles, persistent disequilibrium, and the empirical contradictions in the divergence-of-opinion literature do not require behavioural bias to explain. Repeated upper-tail selection, fed back into subsequent valuations, is sufficient. For the NeurIPS audience the practical consequence is concrete: an ML valuation model trained on historical clearing prices is by construction fitting an upper-order statistic, and selection-aware estimation is a deployment-time requirement, not an academic nicety.

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A ODD Protocol

This appendix documents EDEM in the standard ODD protocol for agent-based models [Grimm et al., 2020].

Purpose and patterns. EDEM characterises the conditions under which an agent-based real-estate market reaches a stable equilibrium price, and identifies the mechanisms that prevent equilibrium when they do not. Patterns the model reproduces: stable band-bounded equilibria; endogenous business cycles; persistent shifts in the realised price relative to the textbook equilibrium under altered patience or density; multiplicative price drift in the absence of a balancer.

Entities, state variables, scales. Entities: Buyers and Sellers (`mesa.Agent` subclasses), Homes (per-cell dataclass on a 32×32 toroidal grid). Buyer state: epsilon (estimation-error bound), delay, current bid, heading; EDEM variant adds the lowest-bid-to-value ratio and a yellow (winning) flag. Seller state: epsilon, patience timer, current ask price, dictionary of received bids; EDEM adds number of bids, total bids, best-bid-to-value, best-bid-to-true-value. Home state: market price, last-sale tick, last-sale price, fair value v^*, current value v. Model state: rolling window of last 25 sale prices (DE), epoch counter (EDEM), cross-population epsilon (EDEM). One spatial unit = one home; one tick is the smallest temporal unit. DE balance period is 100 ticks; EDEM epoch is 20.

Process overview. Per tick: (i) each agent steps once, in randomised order over both classes; a Buyer's step posts bids, then wiggles, then steps forward; a Seller's step processes patience and best-bid logic; (ii) the model invokes the balancer (DE: every 100 ticks; EDEM: every 20 ticks via the cycle-counter trick that preserves buyers' winning-bid flags); (iii) the data collector samples model-level reporters.

Design concepts. *Basic principles*: EDEM extends the divergence-of-opinion premise of Miller [1977] to a multi-period setting; agents act on noisy estimates of an unobservable fair value. *Emergence*: the price level, agent population dynamics, and macroscopic regimes emerge from local interaction; none of these phenomena are coded in directly. *Adaptation*: sellers lower ask prices when patience elapses without high-enough bids (the only adaptive behaviour). Buyers do not adapt. *Objectives*: sellers maximise realised sale price subject to patience. Buyers commit only to offers whose realised bid is at or above their own running benchmark (Eq. (2))—a selection rule, not a price-minimisation objective. *Learning, prediction*: none in the baseline model. *Sensing*: buyers sense whether a seller is on their patch; sellers sense the bids in their own bid table; neither senses the global market price directly. *Interaction*: direct via mirrored-dictionary updates; indirect via the average ask price and rolling sale-price average influencing newly-spawned agents. *Stochasticity*: agent placement, buyer heading, epsilon draws, per-bid error draws, victim selection, Bernoulli draws for fractional C_b swaps—all from a single seeded `numpy.random.Generator`. *Observation*: per-tick model reporters stacked into Parquet datasets, one per seed.

Initialisation. DE: spawn `equi_qnty` sellers at random unique cells with patience $\sim \mathcal{U}[0, \texttt{init\_patience})$ and ask price drawn from $\mathcal{U}[-\bar{\sigma}, +\bar{\sigma}]$ around p^*; spawn `equi_qnty` buyers at random cells. EDEM: 20 sellers and 20 buyers, all homes initialised with $v_0(h) = v^*(h) = 100$.

Input data and submodels. No external input data; EDEM is closed. Estimation function: Eq. (1). Bid acceptance: Eq. (2). Market price: Eq. (3) (DE) and Eq. (4) (EDEM). Balancer: Eq. (5) in finite-population form with population floor of one.

B Parameter Tables

Common parameters across all runs. World size 32×32 toroidal; seeds per run ≥ 8 (typically 10); bid-acceptance rule `netlogo` (the rule actually coded in the original 2018 NetLogo source, matching Eq. (2)).

DE runs 1–5. Supply $\hat{Q}_s(p) = a_s + 0.5\,p$, $a_s = 0$. Demand $\hat{Q}_d(p) = a_d - 0.5\,p$, $a_d = 100$ (Runs 1–3, 5) or 50 (Run 4) or $\{125, 75\}$ alternating (Run 5 Scenario B). Maximum valuation error $\bar{\sigma}$: 0.05

(Runs 1, 3–5) or 0.25 (Run 2). Maximum patience: 50 (Runs 1, 2, 4) or 100 (Run 3) or $50 \to 165$ at $t = 7000$ (Run 5 Scenario A). Balance period $T_B = 100$. Sale window $W = 25$. Ticks per seed: 20,000 (Runs 1–4), 12,000 (Run 5).

EDEM runs 6–8. Initial $\bar{\sigma}$: 0.15 (Runs 6, 7) or 0.05 (Run 8). $\bar{\sigma}$ growth per epoch: 0 (Runs 6, 7) or $+0.005$ (Run 8). Balancer C_b: 0 (Run 6); $\{+1, 0, -1\}$ (Run 7); -1 (Run 8). Initial buyers / sellers: 20/20. Epoch length 20 ticks. Ticks per seed: 3000. True value $v^*(h) = 100$ uniform across all homes.

Run 9 sensitivity grid. $C_b \in \{-1, -0.5, 0, +0.5, +1\}$; $\bar{\sigma} \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. 5 seeds per cell, 1500 ticks each.

C All Experiment Figures

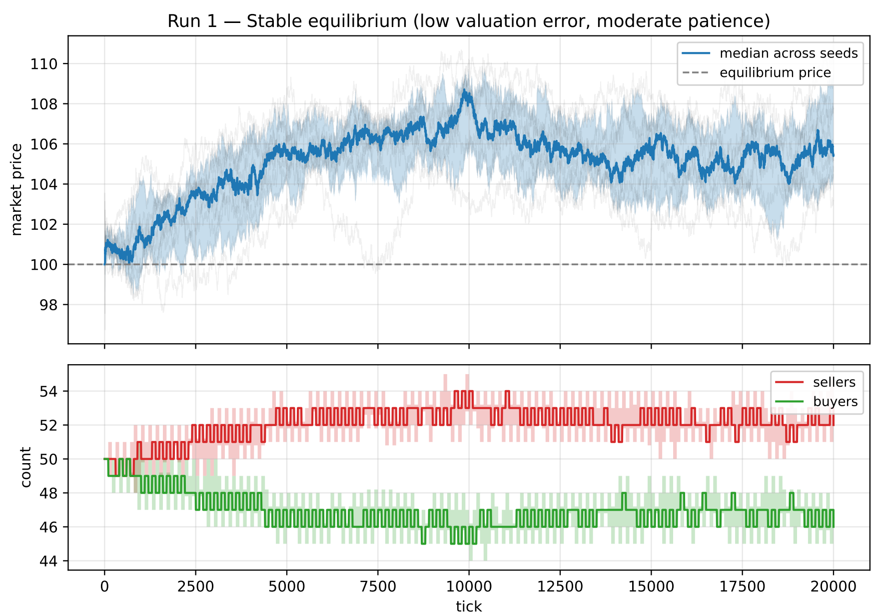

Figure 4: Run 1 – Stable equilibrium with low valuation error and moderate patience.

D Extensions and Future Work

The framework of Section 3 is deliberately minimal so that extensions can be plugged in without rewriting the agent skeleton. Five directions are especially fruitful and we sketch each below.

Bias-corrected clearing. The bid-selection asymmetry embedded in the max-bid clearing rule cannot be cancelled by any linear balancer within the explored grid. Two replacements for the clearing functional A in Eq. (3) would address it directly: (i) a winsorised mean replacing $\bar{r}_t$ in Eq. (4); (ii) a selection-aware estimator substituting the median or lower-percentile of all bids (not only winning bids) for $\bar{r}_t$. Either is a one-line edit to the reference implementation.

Realtors and commission structures. The DE seller is a strict utility-maximiser whose only friction is patience. Real-world residential transactions are intermediated by realtors whose payoff is a fraction of the sale price. A realtor extension would replace the seller agent with an aggregated listing agent whose patience and ask-price strategies optimise expected commission across a portfolio. This is a natural setting for empirical calibration: realtor commission rates and listing durations are publicly observable, unlike the patience parameter of the bare DE seller.

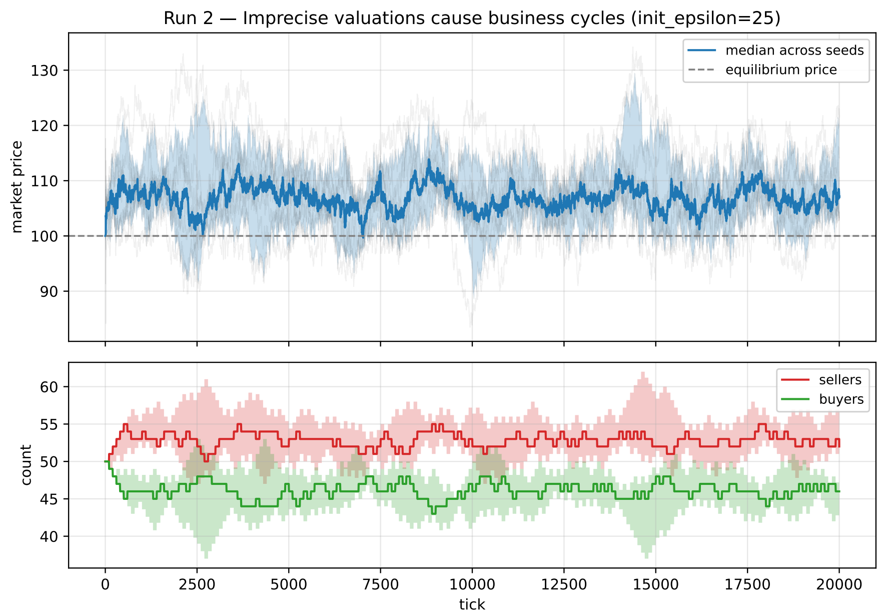

Figure 5: Run 2 – High valuation error ($\bar{\sigma} = 0.25$) generates endogenous business cycles.

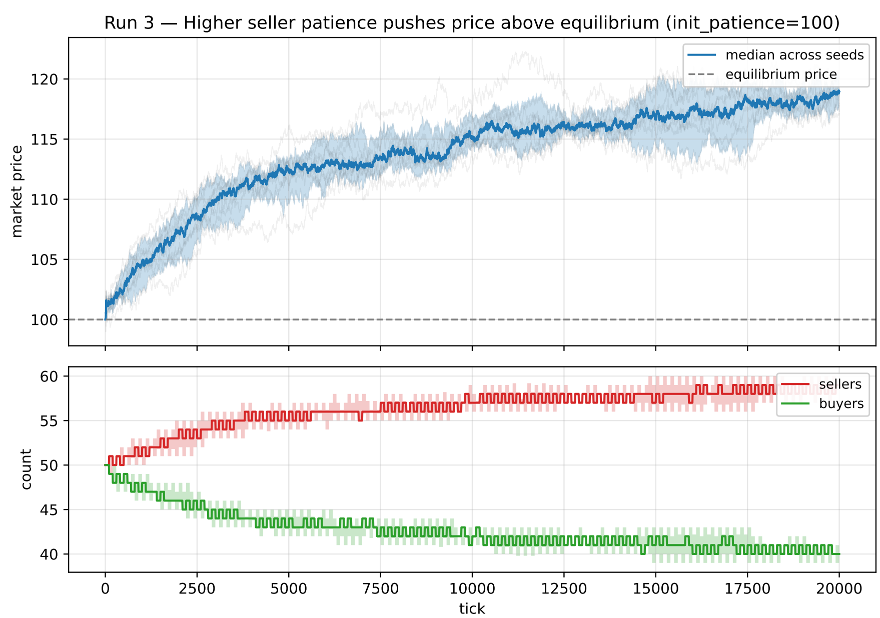

Figure 6: Run 3 – Doubling seller patience to 100 pushes the market price about 14% above the textbook equilibrium and sustains it there.

Open vs. blind auctions. EDEM as posed is a blind-bid model. Open-bid auctions substantially change the bid distribution by exposing buyers to one another's valuations and would likely amplify the bid-selection asymmetry.

Construction and the supply side. A construction-market variant inverts EDEM's agent roles: contractors bid downward to win projects, with the lowest-bid contractor winning. The selection asymmetry then operates in reverse, biasing realised contract prices below the population fair value.

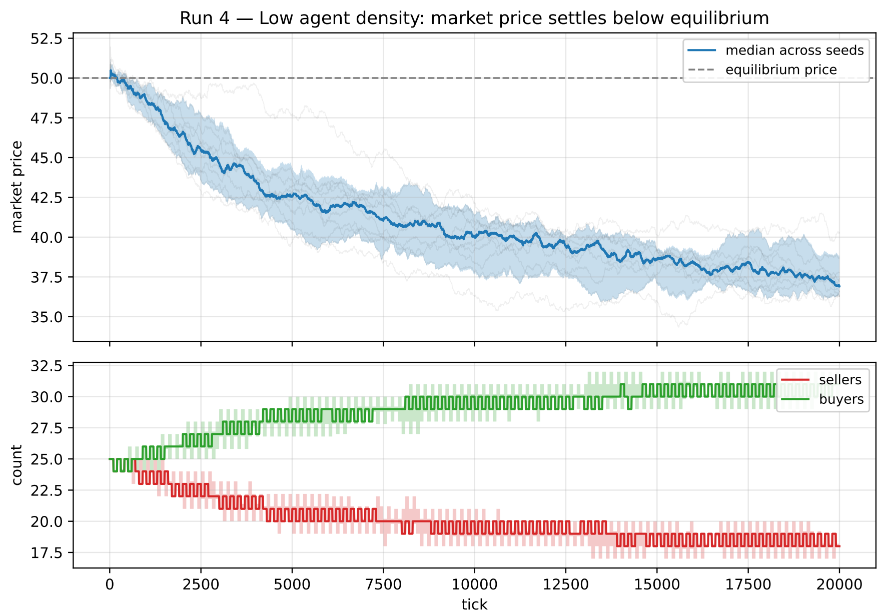

Figure 7: Run 4 – Halving the demand intercept reduces equilibrium quantity to 25; the realised market price settles about 18% below the textbook equilibrium.

Adaptive agents. EDEM agents are non-learning. A Bayesian extension where each agent maintains a posterior over fair value and updates after every observed sale would converge, in the limit, toward bias-corrected estimation. A reinforcement-learning variant with a buy-low/sell-high reward would discover and exploit the bid-selection asymmetry; this would be a computational analogue to the Zillow-Offers iBuying problem [Zillow Group, Inc., 2021] and would offer a stress test of any proposed clearing-rule fix.

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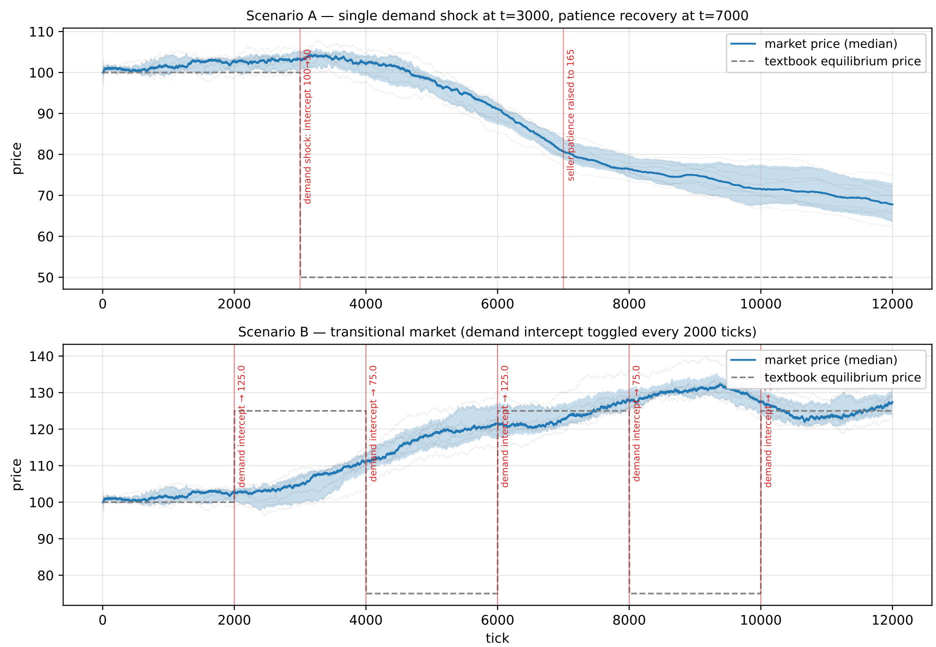

Figure 8: Run 5 – Two shock scenarios. Top: a single demand shock at $t = 3000$ followed by a patience boost at $t = 7000$. Bottom: a transitional market in which demand toggles every 2000 ticks.

Answer: [Yes]

Justification: Section 8 explicitly states the limitations: single asset class, no leverage, no macroeconomic environment, non-adaptive (non-learning) agents, finite parameter grid that does not cover extreme $|C_b|$, and a truncated shock-recovery horizon in Run 5.

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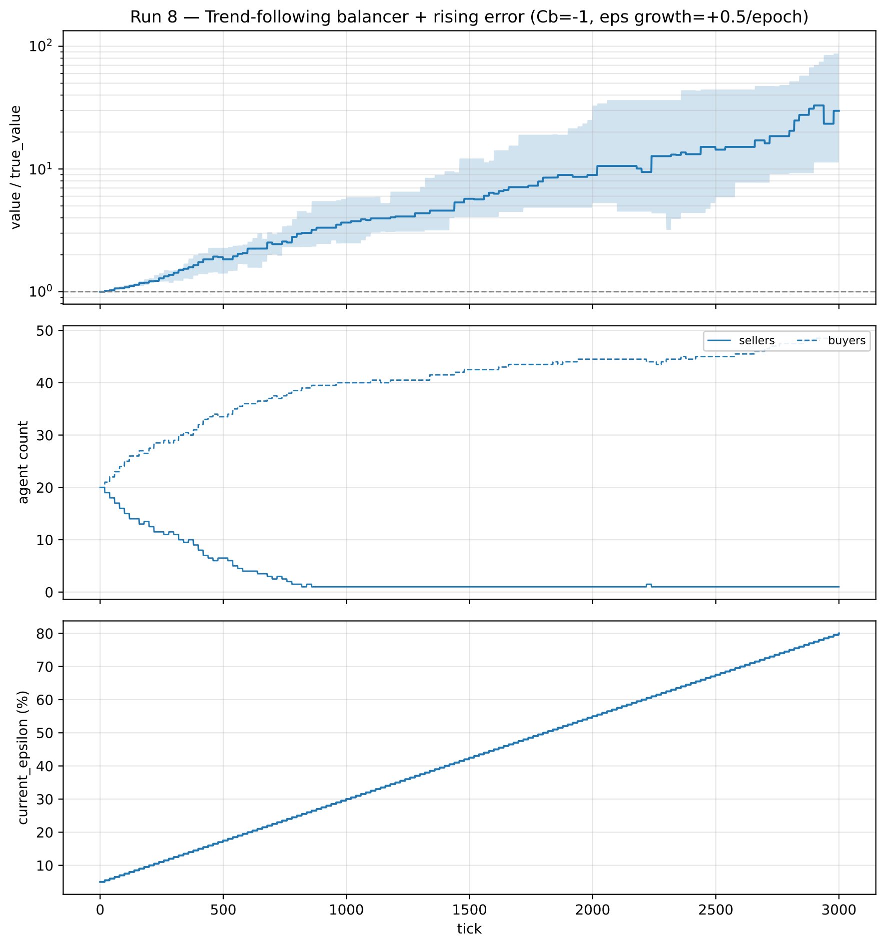

Figure 9: Run 8 – Trend-following balancer ($C_b = -1$) combined with linearly-growing $\bar{\sigma}$ ($0.05 \to 0.80$). The trajectory is sustained super-linear rather than visibly double-exponential.

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